Technique-Guided Coaching Feedback Generation

(a) Offline Checkpoint Design

Technique Description

Parting the Wild Horse's Mane: ending pose

Left bow stance; right leg naturally extended.
Turn the torso slightly to the left.
Left hand at eye level; right hand beside the right hip.

Technique excerpt shown; full text is supplied to the LLM.

Codex

Select measurable posture checkpoints

Fixed at inference

Selected Checkpoints

1 Left knee angle
2 Right knee angle
3 Left elbow angle
4 Right elbow angle
5 Normalized stance length
6 Left wrist height relative to head
7 Left wrist reach relative to head
8 Right wrist-to-hip distance

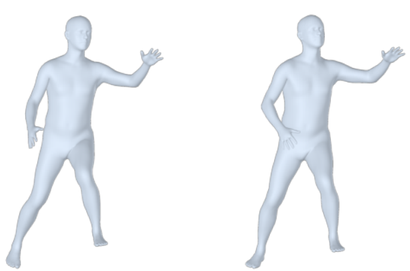
Checkpoint definitions + full technique

(b) Evidence-Based Coaching

Localized SMPL Poses

Parting the Wild Horse's Mane

Reference anchor | Student 10



41.400 s | frame0 1242 | 37.571 s | frame0 1126

Anchor: BV1WE411W7JB

Teacher ranges use 10 teachers.
Only the reference anchor is pictured.

Measurement & Structured Evidence

Native SMPL-24 | wrist offsets normalized by torso length

Checkpoint	Teacher ranges A / B	Student
L knee angle	[64.1, 117.9] [77.8, 117.5]	108.6°
R knee angle	[136.1, 155.0] [137.8, 151.6]	150.5°
L elbow angle	[92.8, 126.5] [100.0, 120.2]	112.0°
R elbow angle	[124.7, 139.6] [127.1, 138.9]	120.8°
Stance length / leg length	[0.542, 0.946] [0.541, 0.860]	0.492
L wrist height vs. head	[-0.089, 0.000] [-0.116, -0.016]	-0.118
L wrist reach vs. head	[0.310, 0.590] [0.305, 0.557]	0.360
R wrist-to-hip / torso	[0.611, 1.062] [0.656, 0.985]	0.673

A: median ± 2 MAD B: P10-P90 (10 teachers)

Student value: median over the preceding 0.2 s + endpoint.

JSON also contains full technique, definitions, center values, window ranges and valid-frame counts.

LLM Reasoning

Text + numeric evidence only

1 Compare every checkpoint

Use both teacher reference ranges.

2 Integrate related evidence

Relate posture to the technique.

3 Assess uncertainty

Consider variability and proxy limits.

Outside range ≠ confirmed error.
Insufficient evidence → abstain.

No video or SMPL image is sent to the LLM.

Coaching Feedback

Example model output (abridged)

Bow stance

Increase step length and deepen the front-knee bend.

Right arm

Let the arm settle naturally beside the hip; avoid excessive elbow flexion.

Left-hand height

Evidence is insufficient; no height correction is suggested.

Coach summary

Focus on a fuller bow stance and a naturally extended right arm.

Technique-guided checkpoints turn aligned pose measurements into evidence-based coaching.